Walmart Weekly Sales Forecasting

A Time Series Analysis & SARIMAX Forecasting Project

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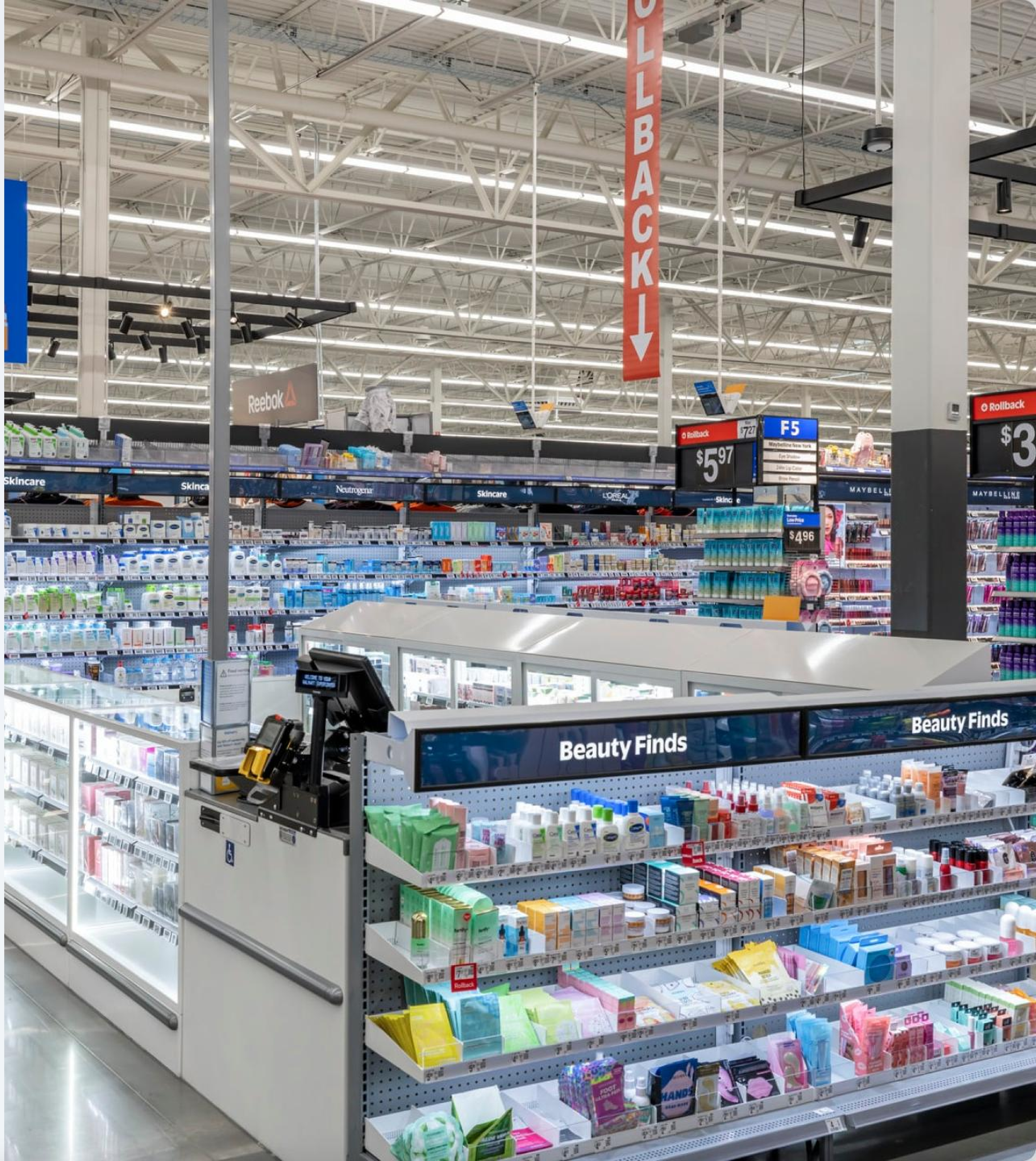
Project Objective

Walmart operates 45 stores across multiple regions, each uniquely influenced by macroeconomic conditions, seasonality, and holiday events.

Goal 1: Analyze historical sales data to map external drivers.

Goal 2: Build a SARIMAX forecasting pipeline for accurate store-level inventory and demand management.



Dataset Overview

Scale & Scope

6,435 weekly records spanning 45 store locations over a 3-year historical period (2010 - 2012).

Key Features

Includes Store, Date, Weekly Sales, Holiday Flag, Temperature, Fuel Price, CPI, and Unemployment rate.

Data Quality

Exceptionally clean and robust dataset with absolutely zero missing values found across all 8 feature columns.

Exploratory Data Analysis

Sales Distribution

Weekly sales exhibit a right-skewed distribution. The vast majority of stores generate below $1.5M per week, while a smaller, elite group of high-performing locations consistently reach well above that baseline.

Temperature Effect

Correlation between weekly sales volume and regional temperature measured at -0.064. This negligible relationship strongly indicates that temperature has virtually no direct influence on consumer spending.

Store Benchmarking

A massive performance gap of ~$1.85M (~87.7%) exists between the top and bottom stores, highlighting substantial disparities in store location, customer base, or store-level execution.

Economic Sensitivity

Economic Indicator	Store Focus	Correlation	Impact Insight
Unemployment	Store 38	-0.785	Sales decline sharply as local unemployment rates rise.
	Store 44	-0.780	
Consumer Price Index	Store 36	-0.915	As the cost of goods rises (inflation), sales volume drops.
	Store 35	-0.424	

Seasonality & Holiday Effects

- **Annual Patterns:** Seasonal decomposition confirms a recurring annual pattern with massive sales spikes.
- **Holiday Concentration:** Spikes are heavily concentrated at the end of each year.
- **Shopping Seasons:** These surges perfectly align with Thanksgiving, Christmas, and New Year shopping events.
- **Variance:** Holiday weeks exhibit measurably higher and more variable sales compared to stable non-holiday weeks.

Busy holiday shopping season

Forecasting Methodology

SARIMAX Model

Configured with order (0,1,1) and seasonal order (0,1,1,52) to properly capture the annual 52-week seasonality inherent in retail.

Train/Test Approach

Series resampled to weekly totals. The last 12 weeks were held out for testing before refitting to generate future 12-week forecasts.

Parallelization

Store-level model fitting was highly optimized and distributed across CPU cores using Joblib to forecast stores concurrently.

Forecast Results (Test)

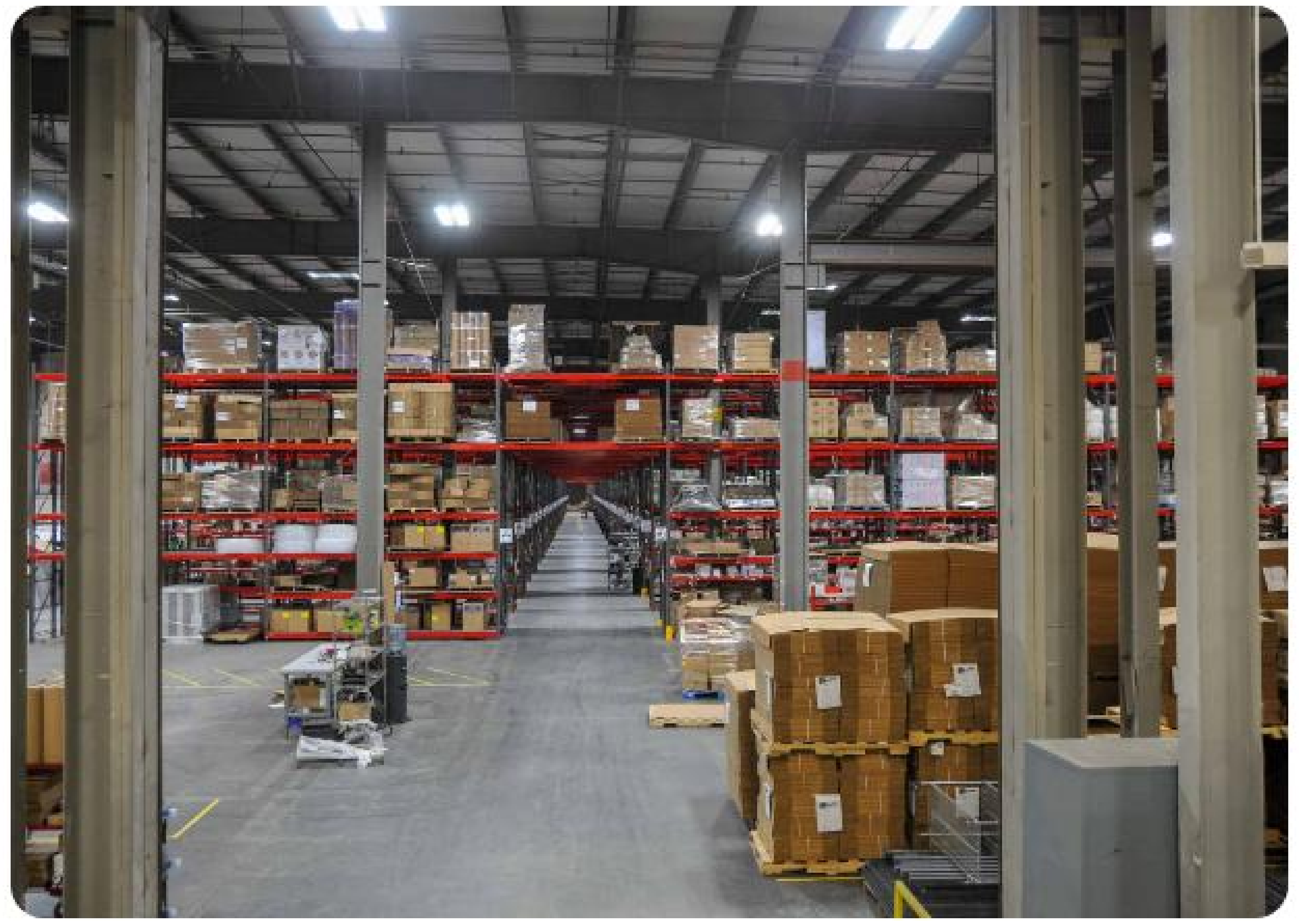
$$\text{MAPE} = \frac{100\,\%}{n} \sum_{t=1}^{n} \frac{A_t - F_t}{A_t}$$

Store ID	RMSE	MAPE (%)	Forecast Accuracy
Store 33	9,140.62	2.97%	High
Store 44	13,192.63	3.13%	High
Store 5	12,240.85	2.61%	High
Store 38	22,128.21	3.65%	High
Store 36	27,825.47	8.38%	Moderate

Key Findings

- **Holiday Dominance:** Holiday seasonality is the single largest driver of sales spikes, tightly concentrated in November and December.
- **Economic Vulnerability:** Economic indicators like unemployment and CPI have a measurable negative effect on a specific subset of stores.
- **Performance Gap:** An ~88% gap exists between top and bottom stores, proving that local and operational factors heavily outweigh macroeconomics.
- **Forecast Reliability:** SARIMAX models achieved ~3-4% MAPE for most stores, confirming historical patterns strongly predict short-term futures.

Strategic Recommendations

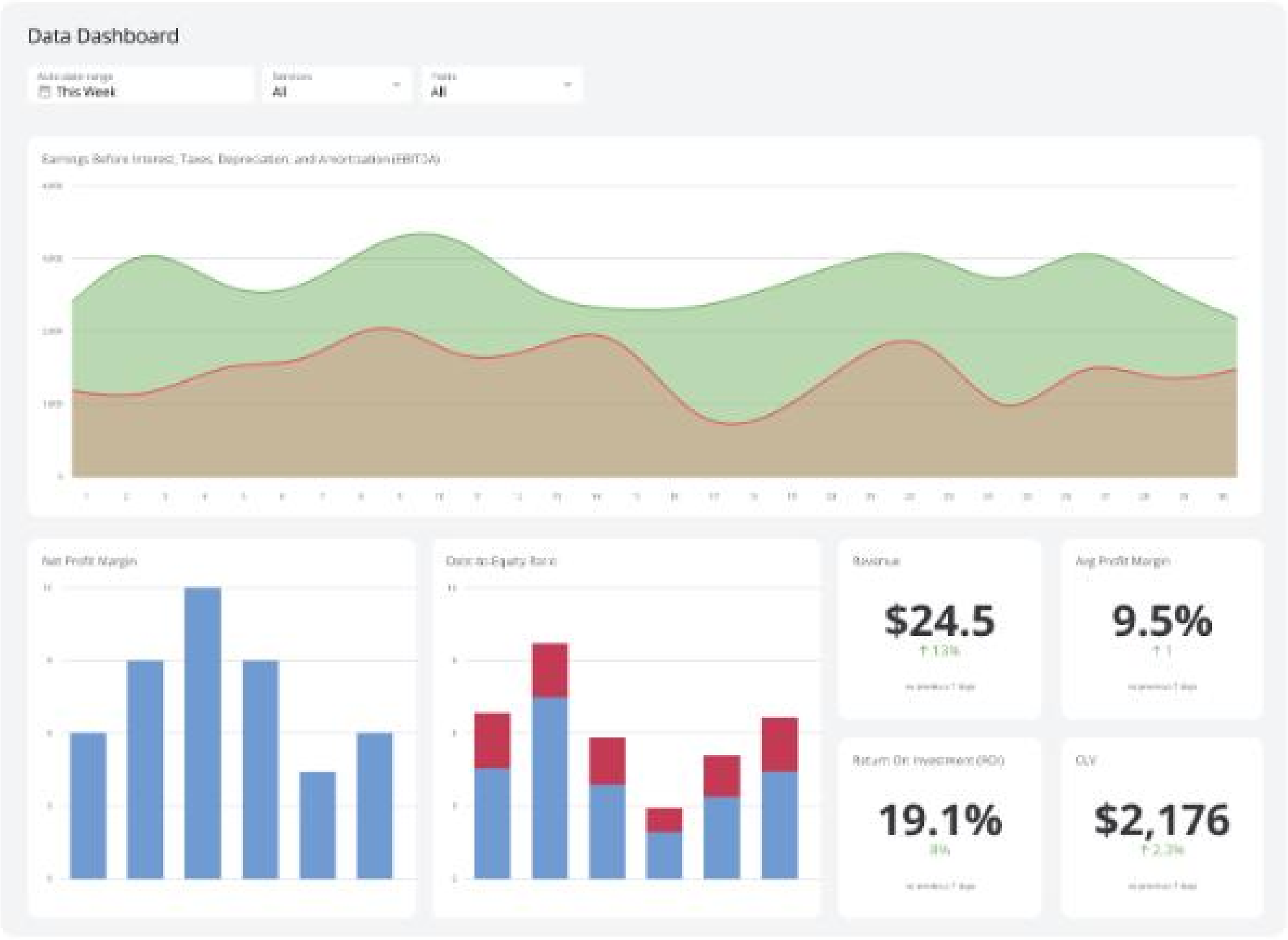


Inventory & Staffing

Prioritize aggressive capacity increases ahead of the Nov-Dec holiday rush across all locations.

Localized Pricing

Implement localized promotions for stores heavily affected by CPI and Unemployment (Stores 36, 38).



Pipeline Expansion

Extend the forecasting pipeline to all 45 stores and refresh models on a rolling monthly basis.

Questions?

Thank you for your attention.

Tools & Technologies Used

Stack: Python, Pandas, NumPy, Statsmodels, Scikit-learn

Parallelization: Joblib (loky backend)

Reporting: XlsxWriter (multi-sheet Excel export)

Image Sources

https://www.brrarch.com/wp-content/uploads/WM_003_N4_website.jpg

Source: www.brrarch.com

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